Ratio Test Problems & Solutions

Infinite Series Convergence Handout

Part 1: Problem List

Test for convergence the following infinite series using D'Alembert's Ratio Test:

1. $x + \dfrac{x^2}{2} + \dfrac{x^3}{3} + \ldots \infty$

2. $\sqrt{\dfrac{1}{2}}x + \sqrt{\dfrac{2}{3}}x^2 + \sqrt{\dfrac{3}{4}}x^3 + \ldots \infty$

3. $1 + \dfrac{x}{2} + \dfrac{x^2}{5} + \dfrac{x^3}{10} + \cdots + \dfrac{x^n}{n^2+1} + \ldots \infty$

4. $\sum_{n=3}^{\infty} \dfrac{x^n}{n(n-1)(n-2)}$

5. $1 + \dfrac{2^p}{2!} + \dfrac{3^p}{3!} + \dfrac{4^p}{4!} + \ldots \infty$

6. $\sum_{n=1}^{\infty} \dfrac{n!\,3^n}{n^n}$

7. $\sum_{n=1}^{\infty} \dfrac{n^3 + a}{2^n + a} \quad (a > 0)$

8. $\sum_{n=1}^{\infty} \dfrac{\sqrt{n}}{\sqrt{n^2+1}}\, x^n$

9. $\sum_{n=1}^{\infty} \dfrac{n^3 - n + 1}{n!}$

10. $\left(\dfrac{1}{3}\right)^2 + \left(\dfrac{1\cdot 2}{3\cdot 5}\right)^2 + \left(\dfrac{1\cdot 2\cdot 3}{3\cdot 5\cdot 7}\right)^2 + \ldots \infty$

11. $1 + \dfrac{1^2\cdot 2^2}{1\cdot 3\cdot 5} + \dfrac{1^2\cdot 2^2\cdot 3^2}{1\cdot 3\cdot 5\cdot 7\cdot 9} + \ldots \infty$

12. $\dfrac{4}{18} + \dfrac{4\cdot 12}{18\cdot 27} + \dfrac{4\cdot 12\cdot 20}{18\cdot 27\cdot 36} + \ldots \infty$

13. $\dfrac{1}{1^p} + \dfrac{x}{3^p} + \dfrac{x^2}{5^p} + \cdots + \dfrac{x^{n-1}}{(2n-1)^p} + \ldots \infty$

14. $\sum_{n=1}^{\infty} \dfrac{3\cdot 6\cdot 9 \ldots 3n}{4\cdot 7\cdot 10 \ldots (3n+1)} \cdot \dfrac{5^n}{3n+2}$

15. $1 + \dfrac{1+\alpha}{1+\beta} + \dfrac{(1+\alpha)(1+2\alpha)}{(1+\beta)(1+2\beta)} + \dfrac{(1+\alpha)(1+2\alpha)(1+3\alpha)}{(1+\beta)(1+2\beta)(1+3\beta)} + \ldots \infty \quad (\alpha, \beta > 0)$